Homework 1

Numerical methods, Practice

The below problems are optional homework exercises related to the topics of the first half of the semester. These may be handed in as announced by the practice teacher (usually uploaded to the Canvas system) with the deadline of the first test. One may gain at most 5 extra points by solving some of these problems. The complete solution of a problem is worth 1 point.

Problems

1. Let the set of machine numbers be $M(5,-4,4)$. Identify the special numbers ε_0, ε_1, M_∞. Map the following numbers into this set:

$$\frac{1}{7},\quad 0.015,\quad 31,\quad 1.68.$$

2. The input error theorem states that—in case of *rounding* to the nearest machine number—the relative error of convertible numbers is bounded by $\frac{1}{2}\,\varepsilon_1$. (This value is usually denoted by ε_M for 'machine epsilon'.) What would be the machine epsilon if *chopping* is applied instead of rounding? Explain!

3. What are the results of the operations

$$\mathrm{fl}(\sqrt{3}) \oplus \mathrm{fl}(\sqrt{5}) \quad \text{and} \quad \mathrm{fl}(\sqrt{5}) \ominus \mathrm{fl}(\sqrt{3})$$

in the set $M(4,-5,5)$? (Start with the square root values rounded to 3 decimal digits.)

4. Prove that for the set of lower triangular matrices

$$\mathcal{L} = \left\{\, L \in \mathbb{R}^{n\times n} \,:\, l_{ij} = 0 \ (i<j) \,\right\}$$

if $L_1, L_2 \in \mathcal{L}$, then $L_1 \cdot L_2 \in \mathcal{L}$, and if L_1 is invertible, then $L_1^{-1} \in \mathcal{L}$.

5. Prove that for the set of lower triangular matrices with entries 1 in the diagonal

$$\mathcal{L}_1 = \left\{\, L \in \mathbb{R}^{n\times n} \,:\, l_{ij} = 0 \ (i<j),\ l_{ii} = 1 \,\right\}$$

if $L_1, L_2 \in \mathcal{L}_1$, then $L_1 \cdot L_2 \in \mathcal{L}_1$, furthermore L_1 is invertible and $L_1^{-1} \in \mathcal{L}_\infty$.

6. Find the inverse using Gaussian elimination in case of matrix $A \in \mathbb{R}^{n\times n}$ with

$$a_{ij} = \begin{cases} 0 & \text{if } i<j, \\ 1 & \text{if } i \ge j. \end{cases}$$

7. Consider the following $n \times n$ tridiagonal matrix:

$$A = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}.$$

Find its LU decomposition using Gaussian elimination. Give an iterative algorithm for the elements of L and U. What is the determinant of the matrix?

8. Show that if A is a 2×2 matrix then

$$\mathrm{cond}_1(A) = \mathrm{cond}_\infty(A).$$

9. Prove that for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$

$$\|\mathbf{a} \cdot \mathbf{b}^\mathsf{T}\|_2 = \|\mathbf{a}\|_2 \cdot \|\mathbf{b}\|_2.$$

10. Verify that for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$

$$\|\mathbf{a} \cdot \mathbf{b}^\mathsf{T}\|_1 = \|\mathbf{a}\|_1 \cdot \|\mathbf{b}\|_\infty \quad \text{and} \quad \|\mathbf{a} \cdot \mathbf{b}^\mathsf{T}\|_\infty = \|\mathbf{a}\|_\infty \cdot \|\mathbf{b}\|_1.$$